PYTHON OPERATORS

BEGINNER-FRIENDLY

Meanings • Examples • Outputs
Easy Hinglish explanations

What is an operator?

An operator is a symbol or keyword that performs an operation on values or variables. For example, **+** adds two numbers and **==** compares two values.

1. Arithmetic Operators

Operator	Meaning	Example	Result
+	Addition - add values	17 + 5	22
-	Subtraction - subtract values	17 - 5	12
*	Multiplication - multiply values	17 * 5	85
/	Division - may return a decimal	17 / 5	3.4
//	Floor division - round down to whole-number quotient	17 // 5	3
%	Modulus - remainder after division	17 % 5	2
**	Exponent - power	2 ** 3	8

Easy memory tip: With 17 divided by 5, the quotient is 3 and remainder is 2. So 17 // 5 = 3 and 17 % 5 = 2.

Quick code example

```
a = 17
b = 5

print(a / b) # 3.4
print(a // b) # 3
print(a % b) # 2
print(a ** 2) # 289
```

2. Comparison Operators

Comparison operators compare two values. Their result is always a Boolean: True or False. These are commonly used in if-else conditions.

Operator	Meaning	Example	Result
==	Equal to	5 == 5	True
!=	Not equal to	5 != 3	True
>	Greater than	5 > 3	True
<	Less than	5 < 3	False
>=	Greater than or equal to	5 >= 5	True
<=	Less than or equal to	3 <= 5	True

Important: = assigns a value (e.g. **age = 18**). == checks whether two values are equal (e.g. **age == 18**).

3. Assignment Operators

Assignment operators store a value in a variable or update its current value.

Operator	Same as	Example / effect
=	Assign	x = 10 → x becomes 10
+=	x = x + value	x = 10; x += 3 → x becomes 13
-=	x = x - value	x = 10; x -= 3 → x becomes 7
*=	x = x * value	x = 10; x *= 3 → x becomes 30
/=	x = x / value	x = 10; x /= 4 → x becomes 2.5
//=	x = x // value	x = 10; x //= 4 → x becomes 2
%=	x = x % value	x = 10; x %= 4 → x becomes 2
**=	x = x ** value	x = 2; x **= 3 → x becomes 8

4. Logical Operators

Operator	Meaning	Example	Result
and	Both conditions must be True	5 > 2 and 8 > 3	True
or	At least one condition must be True	5 < 2 or 8 > 3	True
not	Reverses a Boolean result	not (5 > 2)	False

5. Membership Operators

Membership operators check whether a value exists inside a sequence such as a string, list, or tuple.

Operator	Meaning	Example	Result
in	Value is present	"py" in "python"	True
not in	Value is not present	4 not in [1, 2, 3]	True

6. Identity Operators

Identity operators check whether two variables refer to the same object in memory. For normal value comparisons, use == rather than is.

Operator	Meaning	Example
is	Both names refer to the same object	x is y
is not	They do not refer to the same object	x is not y

Remember: **==** checks equality of values. **is** checks object identity. Use **is None** when checking for **None**.

7. Bitwise Operators (Advanced)

Bitwise operators work on the binary (bit) representation of integers. You can learn these after becoming comfortable with the basic operators.

Operator	Meaning	Example	Result
&	Bitwise AND	5 & 3	1
|	Bitwise OR	5 | 3	7
^	Bitwise XOR	5 ^ 3	6
~	Bitwise NOT	~5	-6
<<	Left shift	5 << 1	10
>>	Right shift	5 >> 1	2

8. Conditional Expression (Ternary)

Choose one value based on a condition in a single line.

```
age = 20
result = "Adult" if age >= 18 else "Minor"
print(result) # Adult
```

Practice Checklist

Try these without looking at the answer first:

1. What are the results of 17 // 5 and 17 % 5?
2. What does 10 == "10" return?
3. If x = 6, what is x += 4?
4. What is the result of True and False?
5. What does "a" in "data" return?

Suggested learning order: Arithmetic → Comparison → Assignment → Logical → Membership/Identity → Bitwise.